

APM1220: Applied Multivariate Data Analysis

FA 1 – Foundations: Matrix Algebra

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Problem 2.2 – Matrix Multiplication

Given

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}.$$

The dimensions are

$$A : 2 \times 2, \quad B : 3 \times 2, \quad C : 3 \times 1.$$

(a) $5A$

Scalar multiplication does not change the dimensions of A .

$$5A = 5 \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}$$

Multiplying every entry by 5,

$$5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

(b) BA

The dimensions are

$$B_{3 \times 2} A_{2 \times 2}.$$

The inner dimensions agree, so the multiplication is defined and the result will be a 3×2 matrix.

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}.$$

Calculate each entry:

$$(BA)_{11} = 4(-1) + (-3)(4) = -4 - 12 = -16,$$

$$(BA)_{12} = 4(3) + (-3)(2) = 12 - 6 = 6,$$

$$(BA)_{21} = 1(-1) + (-2)(4) = -1 - 8 = -9,$$

$$(BA)_{22} = 1(3) + (-2)(2) = 3 - 4 = -1,$$

$$(BA)_{31} = (-2)(-1) + 0(4) = 2,$$

$$(BA)_{32} = (-2)(3) + 0(2) = -6.$$

Therefore,

$$BA = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c) $A'B'$

First calculate the transposes:

$$A' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix}$$

and

$$B' = \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}.$$

The dimensions are

$$A'_{2 \times 2} B'_{2 \times 3},$$

so the result is 2×3 .

Thus,

$$A'B' = \begin{bmatrix} -1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{bmatrix}.$$

Calculate each entry:

$$\begin{aligned}
(1, 1) &= (-1)(4) + 4(-3) = -16, \\
(1, 2) &= (-1)(1) + 4(-2) = -9, \\
(1, 3) &= (-1)(-2) + 4(0) = 2, \\
(2, 1) &= 3(4) + 2(-3) = 6, \\
(2, 2) &= 3(1) + 2(-2) = -1, \\
(2, 3) &= 3(-2) + 2(0) = -6.
\end{aligned}$$

Therefore,

$$A'B' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

Notice that

$$A'B' = (BA)'.$$

(d) $C'B$

First,

$$C' = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix}.$$

The dimensions are

$$C'_{1 \times 3} B_{3 \times 2},$$

so the result is 1×2 .

$$C'B = \begin{bmatrix} 5 & -4 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}.$$

The first entry is

$$5(4) + (-4)(1) + 2(-2) = 20 - 4 - 4 = 12.$$

The second entry is

$$5(-3) + (-4)(-2) + 2(0) = -15 + 8 = -7.$$

Therefore,

$$C'B = \begin{bmatrix} 12 & -7 \end{bmatrix}$$

(e) Is AB defined?

The dimensions are

$$A : 2 \times 2$$

and

$$B : 3 \times 2.$$

For AB to be defined, the inner dimensions must agree:

$$(2 \times \boxed{2})(\boxed{3} \times 2).$$

Since

$$2 \neq 3,$$

the multiplication is not defined.

Therefore,

$$\boxed{AB \text{ is not defined.}}$$

Problem 2.4 – Properties of Inverses

Assume that A^{-1} and B^{-1} exist.

(a) Prove that $(A')^{-1} = (A^{-1})'$

Since A^{-1} exists,

$$AA^{-1} = I.$$

Taking the transpose of both sides gives

$$(AA^{-1})' = I'.$$

Using the property

$$(XY)' = Y'X',$$

we obtain

$$(A^{-1})'A' = I.$$

Also,

$$A^{-1}A = I.$$

Taking the transpose,

$$(A^{-1}A)' = I',$$

so

$$A'(A^{-1})' = I.$$

Therefore, $(A^{-1})'$ is both a left and right inverse of A' . By uniqueness of the inverse,

$$\boxed{(A')^{-1} = (A^{-1})'}$$

(b) Prove that $(AB)^{-1} = B^{-1}A^{-1}$

Consider

$$(B^{-1}A^{-1})(AB).$$

Using associativity,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B.$$

Since

$$A^{-1}A = I,$$

we obtain

$$B^{-1}IB = B^{-1}B = I.$$

Now consider the multiplication in the other order:

$$(AB)(B^{-1}A^{-1}).$$

Again using associativity,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}.$$

Since

$$BB^{-1} = I,$$

we obtain

$$AIA^{-1} = AA^{-1} = I.$$

Thus $B^{-1}A^{-1}$ is the inverse of AB . Hence,

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}}$$

Problem 2.8 – Eigenvalues, Normalized Eigenvectors, and Spectral Decomposition

Given

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}.$$

The eigenvalues are found from

$$|A - \lambda I| = 0.$$

Therefore,

$$\begin{vmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{vmatrix} = 0.$$

Thus,

$$(1 - \lambda)(-2 - \lambda) - 4 = 0.$$

Expanding,

$$-2 - \lambda + 2\lambda + \lambda^2 - 4 = 0,$$

so

$$\lambda^2 + \lambda - 6 = 0.$$

Factoring,

$$(\lambda - 2)(\lambda + 3) = 0.$$

Therefore,

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = -3}.$$

Eigenvector for $\lambda_1 = 2$

Solve

$$(A - 2I)e_1 = 0.$$

We have

$$A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}.$$

Therefore,

$$-x + 2y = 0,$$

so

$$x = 2y.$$

Choose $y = 1$. Then

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Its norm is

$$\|v_1\| = \sqrt{2^2 + 1^2} = \sqrt{5}.$$

Therefore, the normalized eigenvector is

$$e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Eigenvector for $\lambda_2 = -3$

Solve

$$(A + 3I)e_2 = 0.$$

We have

$$A + 3I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

Thus,

$$4x + 2y = 0,$$

which gives

$$y = -2x.$$

Choose $x = -1$, giving $y = 2$. Therefore,

$$v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Its norm is

$$\|v_2\| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}.$$

Hence,

$$e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

The eigenvectors are orthogonal because

$$e'_1 e_2 = \frac{1}{5} \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \frac{-2+2}{5} = 0.$$

Spectral Decomposition

For a symmetric matrix,

$$A = \lambda_1 e_1 e'_1 + \lambda_2 e_2 e'_2.$$

First,

$$e_1 e'_1 = \frac{1}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

Similarly,

$$e_2 e'_2 = \frac{1}{5} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}.$$

Therefore,

$$A = 2 \left(\frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) - 3 \left(\frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right).$$

Thus,

$$A = \frac{1}{5} \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -3 & 6 \\ 6 & -12 \end{bmatrix}.$$

Hence,

$$A = \frac{1}{5} \begin{bmatrix} 5 & 10 \\ 10 & -10 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}.$$

Therefore,

$$\boxed{A = 2e_1 e'_1 - 3e_2 e'_2}$$

where

$$\boxed{e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}}.$$

Problem 2.11 – Determinant of a Diagonal Matrix

Let

$$A = \{a_{ij}\}$$

be a $p \times p$ diagonal matrix, so that

$$a_{ij} = 0, \quad i \neq j.$$

We want to prove that

$$|A| = a_{11}a_{22} \cdots a_{pp}.$$

Using cofactor expansion along the first row,

$$|A| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1p}A_{1p}.$$

Since A is diagonal,

$$a_{12} = a_{13} = \cdots = a_{1p} = 0.$$

Therefore,

$$|A| = a_{11}A_{11}.$$

The cofactor A_{11} is the determinant of the matrix obtained by deleting the first row and first column:

$$A_{11} = \begin{vmatrix} a_{22} & 0 & \cdots & 0 \\ 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{vmatrix}.$$

This is again a diagonal matrix. Applying the same argument repeatedly,

$$A_{11} = a_{22}a_{33} \cdots a_{pp}.$$

Hence,

$$|A| = a_{11}(a_{22}a_{33} \cdots a_{pp}).$$

Therefore,

$$\boxed{|A| = a_{11}a_{22} \cdots a_{pp}}$$

or equivalently,

$$\boxed{|A| = \prod_{i=1}^p a_{ii}.$$

Problem 2.24 – Covariance Matrix

Given

$$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) Find Σ^{-1}

Since Σ is diagonal, its inverse is obtained by taking the reciprocal of every nonzero diagonal entry:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Verification:

$$\Sigma \Sigma^{-1} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

(b) Eigenvalues and eigenvectors of Σ

Because Σ is diagonal, its eigenvalues are its diagonal entries:

$$\lambda_1 = 4, \quad \lambda_2 = 9, \quad \lambda_3 = 1.$$

The corresponding normalized eigenvectors are

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus,

Eigenvalue	Eigenvector
4	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
9	$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
1	$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(c) Eigenvalues and eigenvectors of Σ^{-1}

We have

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\lambda_1 = \frac{1}{4}, \quad \lambda_2 = \frac{1}{9}, \quad \lambda_3 = 1.$$

The corresponding eigenvectors are unchanged:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Problem 2.30 – Means, Covariances, and Linear Combinations

Given

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}, \quad \mu_X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix},$$

and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}.$$

Partition X as

$$X = \begin{bmatrix} X^{(1)} \\ X^{(2)} \end{bmatrix},$$

where

$$X^{(1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad X^{(2)} = \begin{bmatrix} X_3 \\ X_4 \end{bmatrix}.$$

Let

$$A = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}.$$

Partitioning the mean vector,

$$\mu_{X^{(1)}} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \mu_{X^{(2)}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The covariance blocks are

$$\Sigma_{11} = \text{Cov}(X^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

$$\Sigma_{22} = \text{Cov}(X^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix},$$

and

$$\Sigma_{12} = \text{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}.$$

(a) $E(X^{(1)})$

$$\boxed{E(X^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}}$$

(b) $E(AX^{(1)})$

Using

$$E(AX^{(1)}) = AE(X^{(1)}),$$

we obtain

$$E(AX^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Therefore,

$$E(AX^{(1)}) = 4 + 6 = 10.$$

Hence,

$$\boxed{E(AX^{(1)}) = 10}$$

(c) $\text{Cov}(X^{(1)})$

From the upper-left 2×2 block of Σ_X ,

$$\boxed{\text{Cov}(X^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}$$

(d) $\text{Cov}(AX^{(1)})$

Use

$$\text{Cov}(AX^{(1)}) = A \text{Cov}(X^{(1)}) A'.$$

Thus,

$$\text{Cov}(AX^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

First,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix}.$$

Therefore,

$$\text{Cov}(AX^{(1)}) = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 4 = 7.$$

Hence,

$$\boxed{\text{Cov}(AX^{(1)}) = [7]}$$

(e) $E(X^{(2)})$

$$\boxed{E(X^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}$$

(f) $E(BX^{(2)})$

Using

$$E(BX^{(2)}) = BE(X^{(2)}),$$

we obtain

$$E(BX^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The first component is

$$1(2) - 2(1) = 0.$$

The second component is

$$2(2) - 1(1) = 3.$$

Therefore,

$$\boxed{E(BX^{(2)}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}}$$

(g) $\text{Cov}(X^{(2)})$

From the lower-right 2×2 block of Σ_X ,

$$\boxed{\text{Cov}(X^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}}$$

(h) $\text{Cov}(BX^{(2)})$

Use

$$\text{Cov}(BX^{(2)}) = B\Sigma_{22}B'.$$

We have

$$B' = \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

First,

$$B\Sigma_{22} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}.$$

Multiplying,

$$B\Sigma_{22} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}.$$

Therefore,

$$\text{Cov}(BX^{(2)}) = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

Thus,

$$\text{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}.$$

Hence,

$$\boxed{\text{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}}$$

(i) $\text{Cov}(X^{(1)}, X^{(2)})$

This is the upper-right block of Σ_X :

$$\boxed{\text{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}}$$

(j) $\text{Cov}(AX^{(1)}, BX^{(2)})$

For linear transformations,

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = A \text{Cov}(X^{(1)}, X^{(2)}) B'.$$

The dimensions are

$$(1 \times 2)(2 \times 2)(2 \times 2) \rightarrow 1 \times 2.$$

Thus,

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}.$$

First,

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix}.$$

Then,

$$\begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}.$$

Therefore,

$$\boxed{\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}}$$

Problem 2.41 – Mean and Covariance of Linear Combinations

Given

$$\mu_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}$$

and

$$\Sigma_X = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 3I_4.$$

Also,

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}.$$

The dimensions are

$$A : 3 \times 4, \quad X : 4 \times 1,$$

so

$$AX : 3 \times 1.$$

The three linear combinations are

$$L_1 = X_1 - X_2,$$

$$L_2 = X_1 + X_2 - 2X_3,$$

and

$$L_3 = X_1 + X_2 + X_3 - 3X_4.$$

(a) Find $E(AX)$

Using

$$E(AX) = AE(X) = A\mu_X,$$

we obtain

$$E(AX) = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}.$$

For the first component,

$$1(3) - 1(2) = 1.$$

For the second component,

$$1(3) + 1(2) - 2(-2) = 3 + 2 + 4 = 9.$$

For the third component,

$$1(3) + 1(2) + 1(-2) - 3(0) = 3 + 2 - 2 = 3.$$

Therefore,

$$E(AX) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

(b) Find $\text{Cov}(AX)$

For a linear transformation,

$$\text{Cov}(AX) = A\Sigma_X A'.$$

Since

$$\Sigma_X = 3I_4,$$

we have

$$\text{Cov}(AX) = 3AA'.$$

Calculate AA' .

The first row of A is

$$r_1 = [1 \quad -1 \quad 0 \quad 0].$$

The second row is

$$r_2 = [1 \quad 1 \quad -2 \quad 0].$$

The third row is

$$r_3 = [1 \quad 1 \quad 1 \quad -3].$$

The diagonal entries are

$$r_1 r_1' = 1^2 + (-1)^2 = 2,$$

$$r_2 r_2' = 1^2 + 1^2 + (-2)^2 = 6,$$

and

$$r_3 r_3' = 1^2 + 1^2 + 1^2 + (-3)^2 = 12.$$

For the off-diagonal entries,

$$r_1 r_2' = 1(1) + (-1)(1) + 0(-2) + 0(0) = 0,$$

$$r_1 r'_3 = 1(1) + (-1)(1) + 0(1) + 0(-3) = 0,$$

and

$$r_2 r'_3 = 1(1) + 1(1) + (-2)(1) + 0(-3) = 0.$$

Therefore,

$$AA' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

Hence,

$$\text{Cov}(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

Thus,

$$\boxed{\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}}$$

Therefore, the variances are

$$\boxed{\text{Var}(L_1) = 6, \quad \text{Var}(L_2) = 18, \quad \text{Var}(L_3) = 36.}$$

The covariances are

$$\boxed{\text{Cov}(L_1, L_2) = 0,}$$

$$\boxed{\text{Cov}(L_1, L_3) = 0,}$$

and

$$\boxed{\text{Cov}(L_2, L_3) = 0.}$$

(c) Which pairs of linear combinations have zero covariances?

From

$$\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix},$$

all off-diagonal entries are zero.

Therefore,

$$\text{Cov}(L_1, L_2) = 0$$

$$\text{Cov}(L_1, L_3) = 0$$

and

$$\text{Cov}(L_2, L_3) = 0.$$

Hence, every pair of the three linear combinations has zero covariance.

Final Answers Summary

Problem 2.2

$$5A = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

$$BA = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

$$A'B' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

$$C'B = \begin{bmatrix} 12 & -7 \end{bmatrix}$$

$$AB \text{ is not defined.}$$

Problem 2.4

$$(A')^{-1} = (A^{-1})'$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

Problem 2.8

$$\lambda_1 = 2$$

$$e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -3$$

$$e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$A = 2e_1e_1' - 3e_2e_2'$$

Problem 2.11

$$|A| = a_{11}a_{22} \cdots a_{pp} = \prod_{i=1}^p a_{ii}$$

Problem 2.24

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Eigenvalues of Σ :

$$4, 9, 1$$

Eigenvalues of Σ^{-1} :

$$\frac{1}{4}, \frac{1}{9}, 1$$

The corresponding eigenvectors are the standard basis vectors.

Problem 2.30

$$E(X^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$E(AX^{(1)}) = 10$$

$$\text{Cov}(X^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Cov}(AX^{(1)}) = [7]$$

$$E(X^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$E(BX^{(2)}) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$\text{Cov}(X^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

$$\text{Cov}(BX^{(2)}) = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}$$

$$\text{Cov}(X^{(1)}, X^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\text{Cov}(AX^{(1)}, BX^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}$$

Problem 2.41

$$E(AX) = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}$$

$$\text{Cov}(AX) = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

All three pairs have zero covariance:

$$\text{Cov}(L_1, L_2) = 0, \quad \text{Cov}(L_1, L_3) = 0, \quad \text{Cov}(L_2, L_3) = 0.$$